**Title**

# Delay Coordinate Embedding as Neuronally Implemented Information Processing: The State Space Theory of Consciousness

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**Abstract:**

Despite intensive effort, a satisfactory scientific theory of consciousness remains elusive. This paper introduces the State Space Theory, aiming to meet key criteria for a theory of consciousness while providing a satisfactory framework consistent with neurobiological and philosophical considerations. The theory posits that the cortex processes information through delay coordinate embedding within recurrent neural networks. These networks create state space representations of reality, forming hierarchical, pseudo-hierarchical, and parallel pathways. Under this theory, consciousness is posited to arise at the highest-order engines in this hierarchy, with complex behavioral options competing within these engines. The theory emphasizes that consciousness is a dynamic process, not representable within a static neuronal state, and develops uniquely in each individual due to history-dependent development of the processing engines.

Neuronally, these delay coordinate embedding engines are posited to form recurrent networks that process information by integrating current and past data to reconstruct models of reality, leveraging the power of Takens' Theorem. These engines would themselves exhibit nonlinear dynamics sensitive to initial conditions. This mechanism aligns with the subjective and emergent properties of consciousness, explaining phenomena like binocular rivalry and ambiguous figures. The theory also addresses cortical plasticity and the heuristic nature of cortical processing, aligning with neurobiological evidence supporting the idea that the cortex's information processing is generic rather than specialized.

The State Space Theory aligns with and expands upon major theories like Higher-Order Theories, Global Workspace Theories, Integrated Information Theory, and Predictive Processing Theories by providing a computational mechanism for hierarchical integration and recurrent processing. Philosophically, it reconciles dualist intuitions with a monist perspective, positing that consciousness emerges from dynamic brain processes rather than static states. The theory addresses the unity of consciousness and the privacy of subjective experiences, offering a new framework to understand free will within deterministic systems. Future work will refine this theory, exploring its neural mechanisms and validating its predictions, advancing our understanding of consciousness.

**Key Words: T**heory of Consciousness, Delay Coordinate Embedding, Recurrent Neural Networks, Hierarchical Integration, Nonlinear Dynamics

## Introduction

Despite a substantial effort, which has dramatically increased over the past several decades, a satisfactory scientific theory of consciousness remains elusive. Spectacular advances in our ability to study the behavior of neuronal systems *in situ* and at scale have not yet yielded the theoretical bridge that connects the behavior of these systems with an explanation of phenomenal consciousness; as yet, the hard problem of consciousness has not been solved even as proffered physicalist theories of consciousness continue to multiply.(Seth & Bayne 2022)

The problems that continue to plague theories of consciousness have been well described; a recent paper by Doerig and colleagues offers a reasonable starting point for assessing the viability of these and proposed theories.(Doerig et al. 2021) In it, the authors identify that theories must be constructed in such a way as to cope with a number of key questions and criteria. These elements represent a summary of the major neuroscientific, analytical, and philosophical considerations and quandaries affecting the viability of various candidate theories of consciousness.

## Presentation of the Theory

This paper aims to develop a theory of consciousness that aims to fulfill these and other key criteria. The main ideas put forth in this paper - the State Space Theory - are as follows:

1. Generic computation - The computational processing that occurs in the cortex is somewhat generic rather than highly specialized. This is evidenced through neuroscientific observations including interindividual variability in normal brain function (primary visual cortex; Broca's area) as well as plasticity in response to congenital

disorders, trauma, or surgery (visual cortex activation in Braille reading; hemispherectomy).

2. Delay coordinate embedding - The specific approach taken by the cortex to process information occurs through a *process* called delay coordinate embedding. Short- and long-range recurrent cortical networks form the neurobiological basis for delay coordinate embedded processing in the cortex. Neuronally implemented delay coordinate embedding engines are responsible for the information processing that results in the instantiation of representations of reality within neuronal systems.
3. Recurrent networks are not unfoldable - Under the hypothesis, delay coordinate embedding engines are modified in the process of firing, allowing engines to both learn and represent different aspects of the state space being represented. In addition, delay coordinate embedding is a *process* rather than an *algorithm*. As a result, these recurrent processing engines would not fulfill criteria necessary to unfold a recurrent network into a feedforward network.
4. State space representations - Delay coordinate embedding creates representations of reality by encoding aspects of reality positioned within a state space describing that aspect of reality.
5. Hierarchical organization - Delay coordinate embedding engines are arranged pseudo-hierarchically, with higher order delay coordinate embedding engines representing increasingly abstract and synthesized aspects of reality. While there is a hierarchical organization, multiple parallel processing pathways also exist, consistent with observational neuroscientific data (e.g. dorsal and ventral visual pathways).

6. Consciousness at the apex - Consciousness arises at the highest-order delay coordinate embedding engines in this hierarchy, with multiple parallel highest-order delay coordinate embedding engines competing to give rise to different complex behavioral options available to the conscious entity.
7. Consciousness as process - Given that delay coordinate embedding is a process that requires signal processing to occur over time, neuronally implemented delay coordinate embedding engines *require* that information processing occur over time in order to represent aspects of reality. This implies that consciousness occurs as a process and cannot be represented within a static neuronal state.
8. History-dependent development - The specifics of how each individual brain processes information is determined in a history-dependent manner. The instantiation of specific aspects of reality within an individual's brain, even at the earliest level of cortical processing, are similar but not identical. Thus, each individual brain instantiates their representation of reality in a similar but unique manner. This results, ultimately, in qualia and consciousness being private and non-shareable.
9. Nonlinear dynamics - The delay coordinate embedding engines themselves are nonlinear dynamical systems sensitive to initial conditions when performing state space categorization. A given representation of reality is selected based on initial conditions. This provides a physiologic basis for commonly observed phenomena such as binocular rivalry and perception of ambiguous figures such as the Necker cube.
10. Limbic modulation - Under the State Space Theory, the highest order delay coordinate embedding engines are modulated diffusely by limbic projections, giving rise to emotional qualia as well as affecting decision-making.

Together, these ideas form the core of the State Space Theory. The overarching hypothesis is that the cortex employs delay coordinate embedding to process sensory information, creating a hierarchy of state space representations that give rise to consciousness at the highest levels of the hierarchy. Crucially, this process is generic, occurs over time, develops in a history-dependent manner, and exhibits nonlinear dynamics.

**Neuronal Firing as Time Series Data**

Leaving aside panpsychist or quantum theories of consciousness, the core informational unit of neuronal systems is the action potential.(Byrne 2023) In response to sensory input, neurons adjust their firing patterns, achieving firing rates approaching 1000 Hz but more typically with an upper bound in the 200 Hz range.(Wang et al. 2016) What is less clear is the computational mechanism that transforms these neuronal firing patterns into coherent perceptual experiences and higher-order cognitive phenomena. This has yet to be defined even within simple model systems such as the 302-neuron *Caenorhabditis elegans.(Hobert 2018)* Powerful modern machine learning approaches have been used to identity what neurons in these systems are doing, but have yet to identify how.(Atanas et al. 2023; Chang et al. 2023)

Let us begin with the view that neuronal action potentials represent time series data. Though discretized into binary signals (firing/not firing), it is well established that summation through excitation and inhibition occurs at the dendritic level, resulting in the ability of neurons to process information nonlinearly.(Payeur et al. 2019; Larkum et al. 2018)

It is into this context that I introduce a mathematical technique commonly used in the analysis of complex systems. This technique, called delay coordinate embedding, is used for attractor reconstruction in these systems. In 1981, Floris Takens demonstrated that the nonlinear dynamics of a strange attractor with a state space (phase space) of $n$ dimensions can be reconstructed from a single observed variable using time-delayed copies of that variable as surrogates for the unobserved dimensions.(Takens 1981)

More formally, given a time series $x(t)$, a delay embedding can be constructed as a set of vectors $y(t) = [x(t), x(t-\tau), \ldots, x(t-(m-1)\tau)]$ where $\tau$ is a fixed time delay and $m$ is the embedding dimension. Under Taken's Theorem, with appropriate choice of $\tau$ and $m$, the reconstructed attractor $y(t)$ preserves the topological and dynamical properties of the original system.

The power of delay embedding lies in its ability to uncover hidden structure and determinism in seemingly complex or chaotic data. It has found wide application in fields ranging from physics and engineering to economics and climatology.(Kantz & Schreiber 2003) This deep and important theorem for the analysis of nonlinear dynamical systems has a seemingly trivial and trite use case in the analysis of simple dynamical systems. Take the example of a car moving on a two-dimensional circular racetrack, moving along dimensions $(x,y)$. Let us say that all we have available to us is a measure of the velocity of that car in one dimension $\bar{v}_x$. As a function of time, $\bar{v}_x(t)$ takes the form of a sine wave (Figure 1, left panel). If we were to develop a delay embedding reconstruction of the behavior of this system using embedding dimension $m = 2$, a trivial exercise to perform, we would reconstruct the motion of the car; the reconstructed attractor of this system is a simple circle (Figure 1, right panel). The reconstruction of the far more complex Lorenz attractor from a single observed dimension is also demonstrated (Figure 1, bottom).

Thus, delay coordinate embedding is also a powerful tool for reconstructing and predicting the behavior of well-defined deterministic systems. The conceptual connection being made in this paper is that the reconstruction and prediction of deterministic systems, a primary function of the brain, is performed in the brain through dendritic integration of data via recurrent neural loops. Under this theory, recurrent networks in the brain represent a neuronal implementation of the delay coordinate embedding process. It is proposed that the brain's ability to construct meaningful representations from the raw time series data of neuronal firing patterns occurs through a process that is mathematically described by delay coordinate embedding.

Specifically, the hypothesis being presented is that cortical microcircuits, centered around cortical columns, act as delay coordinate embedding engines. The recurrent connectivity within these microcircuits, spanning cortical layers and involving both excitatory and inhibitory neurons, provides the necessary architecture for integrating information by causing information (spiking activity) that is currently being presented to the system to be integrated with information that had previously been presented to the system via recurrent loops. A vastly simplified diagrammatic representation of a neuronally implemented delay coordinate embedding process is shown in Figure 2. This diagram, while instructive, is vastly simplified in several ways. In actual cortical networks, neurons have thousands or tens of thousands of synapses per dendrite, with a mixture of glutamatergic and cholinergic excitatory synapses, inhibitory GABAergic synapses, and synapses of other neurotransmitter types as well.(DeFelipe & Fariñas 1992) The diagram lays out a view of the reconstruction of an attractor in low ($m = 3$) dimensional space using only one observed variable. Given the large amount of simultaneous information available to the cortex and the requirement for multiplexing of that information to develop a coherent reconstruction of reality, state space reconstruction of the cortex must be multivariable and is likely high dimensional as well (see e.g. Han *et al* for known approaches to multivariable reconstruction of state spaces using nonuniform embedding).(Han et al. 2019)

There is a neuroanatomical basis for the implementation of delay coordinate embedding in the cerebral cortex. It is well established that there are a large number of feedback connections in the cerebral cortex, and that these feedback connections have important functional roles. Specific macroscopic loops include the cortico-basal ganglia-thalamo-cortical loop (Baladron & Hamker 2023; Parent & Hazrati 1995), the cortico-cerebellar loop (Ramnani 2006), the limbic loop (Papez's Circuit) (Shah et al. 2012), the oculomotor loop (Perrinet et al. 2014; Smith & Archibald 2020), and prefrontal cortex loops.(Badre & Nee 2018; Fuster 2015) Dysfunction of or interruption to these recurrent loops result in a wide variety of neurological and psychiatric disorders, including depression, bipolar disorder, and schizophrenia.

Cytoarchitecture at the cortical column level also supports local recurrent loops consistent with a delay coordinate embedding neural analogue. A cortical column is a vertically organized group of neurons within the cortex. The idea of the cortical column was first introduced by Vernon Mountcastle in the 1950s.(Mountcastle 1957; Wikipedia contributors 2023) Mountcastle's work showed that neurons in a given vertical column of the cortex, spanning all six layers, often respond to similar types of stimuli. This observation led to the idea that the cortical column serves as a fundamental unit of information processing in the brain. Cortical columns typically contain about 10,000 neurons, although the exact number can vary across different species and cortical regions. In humans, estimates vary from 100,000 total columns in the neocortex to upwards of 2 million.(Landing et al. 1998; Krueger et al. 2008) Within a column, so-called minicolumns are organized, with short-range connections between them. These neurons are interconnected in complex ways, forming both local loops within the column and connecting to other columns and brain regions.(Schubert et al. 2007; Fox 2018)

There are feedback connections to subcortical structures such as the thalamus and hippocampus as well. These abundant feedback connections in the cortex are essentially delivering old information back to the same dendritic tree that is receiving current information, after a delay. The ability of old information to interact with the newer information at the level of the dendritic tree has clear analogies to delay coordinate embedding. Under this hypothesis, the representation of reconstruction space is spread out over the neural network, with subnetworks representing various parts of that reconstruction space. The recurrent networks, and coincidence detection occurring in the dendritic tree, would represent accessing various parts of reconstructed space. Furthermore, the synaptic weights would be continuously (albeit slowly) updated over time, resulting in learning by the neural network. Though dendritic integration can occur over hundreds of milliseconds, it is more typical for this integration to occur on the scale of tens of milliseconds, corresponding to a frequency of 40-80Hz -- precisely the beta and gamma range frequencies that are known to be associated with conscious experience.(Kirch & Gollo 2020; Purdon et al. 2015)

Under this hypothesis, the neurobiological details underlying the ability of recurrent neural networks to first learn and then access reconstructions of reality would involve a complex interplay between cholinergic and glutamatergic signaling (on the excitatory side) and

GABAergic signaling (on the inhibitory side); the formation and pruning of new synapses; and the modulation of this signaling by myriad other widely-projecting neurotransmitter systems (primarily, dopaminergic, serotonergic, and peptide systems). The specific neurobiology of the development of these systems has and continues to be the subject of deep study across the neuroscientific community. The State Space Theory offers a framework by which to understand the overall trajectory by which the brain reconstructs models of reality and gives rise to conscious experience.

This hypothesis is consistent with observations of neuropsychiatric conditions that arise from demyelinating disorders such as multiple sclerosis. Myelin is a fatty covering found on the axons of some neurons, and facilitates a process called saltatory conduction which dramatically accelerates the speed of the action potential along the length of the axon. Demyelination, then, would be expected to reduce the speed of signals along the axon and disrupt delay coordinate embedding processes.

Analogously, any process that disrupts high-frequency components of the delay embedding processes - particularly those that would act across broader areas of the cortex - would be expected to cause disruption of consciousness. Thus, the State Space Theory offers a long-sought final common pathway for sedation and general anesthesia. Under this theory, general anesthesia can be induced in a variety of ways - through action at GABA receptors, NMDA receptors, or through disruption of cellular energetics and result impacts on neurotransmitter endocytosis.(Weir 2006; Jung et al. 2022; Lambert 2020)

## State Space Exemplars

### *Color Vision*

In standard trichromatic human vision, three retinal types of cone cells are responsible for mediating color vision at the level of input signal to the eye. These cone cells contain pigments sensitive to short, medium, and long wavelength visual light, corresponding to the S-, M-, and L-type classification of these photoreceptive cells.(Carroll & Conway 2021) From a state space perspective, color (hue) can be seen as a state space in three dimensions (Figure 3, top). Certainly, this is a naive perspective that does not take into account the known aspects of color processing that occurs at multiple levels, beginning as early as the retina and continuing through early visual cortex up (V1) through higher-order cortical areas responsible for visual processing (V3, V4). Regardless, a state space perspective on color vision is, at minimum, an informative and useful way to conceptualize the dimensions of known color vision. For example, loss of the M-type photoreceptor results in deuteranopia, the most common form of color deficiency. Figure 3 (bottom) provides a state space representation of color accessibility for deuteranopia. Under the State Space Theory, the qualia for the color space available to deuteranopes remains private. Many attempts have been made to simulate the experience of humans with color deficiency but it should be clear that these are just simulations rather than a true representation of the qualia of those with color deficiencies.

It is clear from observational data that color vision appears to be modular in the building of the representation of objects in perception. Oliver Sacks famously describes the case of Jonathan I in "The Case of the Colorblind Painter."(Sacks 1995) As a result of damage to areas on the ventral occipital cortex, subjects with cerebral achromatopsia have complete loss of color vision.(Bouvier & Engel 2006) This was described by Jonathan I in stark terms - everything "utterly gray and void of color… not just that colors were missing, but that what he did see had a distasteful, 'dirty' look, the whites glaring, yet discolored and off-white, the blacks cavernous — everything wrong, unnatural, stained, and impure."

This is consistent with the State Space Theory, which posits that the brain constructs perceptual experiences by navigating and integrating multidimensional state spaces. Damage to specific cortical areas disrupts the neural processes that construct the 'color' state space, resulting in the specific perceptual deficit of achromatopsia without concomitant loss of other perceptual capabilities (assuming brain damage is restricted to just color processing). The modular nature of color processing is reflected in how the brain segregates and processes different dimensions of visual information, feeding it forward into the construction of a coherent perceptual experience based on an integrated combination of lower order state spaces.

*Motion Perception*

Motion perception provides another exemplar of the State Space Theory. The brain's ability to perceive motion relies on integrating temporal sequences of visual information, constructing a state space that represents dynamic changes in the environment. Neurons in the medial temporal (MT) area of the visual cortex are specifically tuned to detect motion direction and speed.(Solomon et al. 2011) These neurons process sequential visual inputs, integrating them into a coherent perception of motion.

The State Space Theory explains motion perception as the brain's ability to reconstruct state spaces that capture temporal changes in visual stimuli. This dynamic state space allows the brain to predict and interpret motion, integrating information over time to create a unified perception of moving objects. Under the State Space Theory, it is the recurrent loops and feedback connections within the visual cortex that facilitate this integration, serving as the computational mechanism and explanation by which univariate neuron firing results in the ability of the MT area (in general) to discriminate motion. Using purely one-dimensional data, it would (under Taken's Theorem) be expected that the neuronal network would be making use of an embedding dimension of at least 7 to reconstruct the behavior of an object moving through three dimensional space (embedding dimension $m$ is calculated as greater than or equal to twice the underlying dimensionality $d$ plus one).(Takens 1981) As the brain makes use of multichannel data, it is possible that the neuronal network uses a smaller embedding dimension.

*Depth Perception*

Depth perception, the ability to perceive the world in three dimensions, also aligns well with the State Space Theory. The brain uses various cues, such as binocular disparity, motion parallax, and texture gradients, to construct a three-dimensional state space that represents depth.(Howard 2012) Neurons in the primary visual cortex (V1) and higher visual areas, such as the dorsal stream, process these cues to create a coherent perception of depth.

The State Space Theory posits that depth perception arises from the brain's ability to integrate multiple sensory inputs into a unified state space. This integration involves reconstructing spatial relationships between objects, allowing the brain to interpret and navigate the three-dimensional environment. Under the theory, the hierarchical processing of depth cues, from V1 to higher-order visual areas, supports the construction of a multidimensional state space that represents the spatial structure of the environment. This information would then be joined with other state space representations to create a representation that binds e.g. object, color, motion into a unified percept in a higher order delay coordinate embedding engine.

## Implications of Plasticity for Cortical Computation

There are some key constraints about the way that information must be processed by the brain related to plasticity, genetic limitations, and robustness of the network. The mammalian cortex exhibits a striking degree of gross scale plasticity. This plasticity is classically exemplified by experiments describing the patterning of ocular dominance columns observed by Wiesel and Hubel in kittens.(Berardi et al. 2003) In these experiments, Hubel and Wiesel discovered that eye closure during the first six months of life led to a functional, permanent disconnection between the closed eye and the cortex, even though the retina of the closed eye remained intact. Furthermore, the functional eye was preferentially represented in the cortex. This was the first demonstration of Hebb's Rule: the "fire-together, wire-together" principle.

Another example is the role of occipital cortex in Braille reading. Functional MRI and rTMS studies have together established that regions of the cortex that areas normally devoted to visual processing are, in fact, recruited in congenitally blind subjects to subserve the reading of Braille.(Pascual-Leone & Torres 1993; Kupers et al. 2007) Massive cortical reorganization and plasticity have also been noted in sighted subjects who underwent an intensive Braille-reading training regimen.(Siuda-Krzywicka et al. 2016)

Interindividual variability in otherwise healthy subjects also demonstrates the deep role that plasticity plays within the context of normal development. In an fMRI study of subjects viewing grating patterns, Kamitani and Tong were able to train a simple machine learning algorithm to predict the grating that a subject was observing using only fMRI data from visual cortex.(Kamitani & Tong 2005) However, an algorithm trained on one subject could not be used to predict on another subject. Why? Even in the early primary visual cortex, there is substantial interindividual variability in the specific pattern of cortical areas that are sensitive to specific grating orientations.

These and many other examples amount to a well-established plethora of evidence of plasticity found within the mammalian cortex. The presence of plasticity - this ability of the brain to recruit unused or underused neuronal circuitry to subserve new and, in some cases, completely different processing tasks - has important implications for the way that information processing must occur in cortex.

Specifically, it implies that cortical mechanisms of information processing are, at least to some extent, generic. Returning to observations related to Braille-reading patients, the rTMS study in particular underscores the point that the occipital cortex is not merely recruited to fire in the presence of tactile stimuli, but is in fact functionally important for the reading of Braille.(Kupers et al. 2007) The circuits involved in the analysis of the incoming visual data were somehow recruited to functionally process incoming tactile data instead. If cortical information processing systems were highly specialized to subserve particular tasks, this would not be possible.

In addition to plasticity, the size of the human genome also strongly suggests that cortical information processing must be heuristic in nature. There are only about 20,000 protein coding genes in the human genome, far fewer than previously expected (this number goes up when accounting for splice variants, but is on the same order of magnitude).(Nurk et al. 2022) This relatively small number of genes makes it vanishingly unlikely that the blueprints for highly specialized neural circuitry for the processing of visual, auditory, tactile, proprioceptive, linguistic, and other types of data are hardcoded into the genome. Indeed, it is far more likely that the genetic specification of cortical structure gives rise to a system that is able to 'pull itself up by its own bootstraps' to learn the rules of reality and perform any other processing tasks important for survival.

Delay coordinate embedding and the State Space Theory provide just such a computational platform to explain this generic yet highly adaptable nature of cortical processing. The concept of delay coordinate embedding, rooted in chaos theory and complex systems analysis, allows the brain to dynamically reconstruct state spaces from temporal sequences of neural inputs. This reconstruction process involves recurrent loops and feedback mechanisms that integrate past data with current sensory inputs, providing an explanatory framework for learning and adaptation in a continuously evolving neural environment, including adaptability and recruitment of brain areas to subserve new tasks in the context of congenital disorders or trauma.

**Nonlinear Dynamical Computational Engines**

It was mentioned in the introduction that delay coordinate embedding engines are themselves nonlinear dynamical deterministic systems. This aspect merits a bit more expansion, as it is a key feature that arises from the State Space Theory and provides additional explanatory power for crucial aspects of our intuition of consciousness.

As outlined above, neuronally implemented delay coordinate embedding engines are comprised of neural networks that engage in recurrent processing, with integration of time delayed data occurring at the level of the dendrite. The complexity of this recurrent processing and time-delayed data integration renders these systems nonlinear and dynamical. This means that the state of the system at any given moment is influenced not just by the immediate inputs, but also by the history of past inputs and the interactions within the system. Nonlinear dynamical systems are known for their sensitivity to initial conditions, leading to phenomena such as chaos and bifurcations.

The nonlinearity and sensitivity to initial conditions provide a basis for understanding some of the more mysterious aspects of consciousness. For example, the subjective experience of consciousness, or qualia, can be seen as emerging from the complex interactions and feedback loops within these nonlinear systems. The same input can lead to different conscious experiences depending on the prior state of the system and the exact dynamics at play. To return to the example of ambiguous figures mentioned in the introduction, such as the Necker cube or Rubin's vase, these perceptual phenomena can be explained by the brain's nonlinear dynamical processing. When presented with such ambiguous stimuli, the brain does not have a single, static representation but rather multiple potential interpretations that compete with each other (Figure 4). The final conscious percept that emerges can shift between these interpretations depending on subtle changes in the neural network's state or initial conditions, due to the sensitivity to initial conditions characteristic of nonlinear systems.

Another critical implication of viewing delay coordinate embedding engines as nonlinear dynamical systems is their capacity for producing emergent properties. Emergence occurs when a system exhibits properties or behaviors that are not evident from the sum of its parts. In the context of consciousness, this means that the intricate interplay of numerous simple neural components can give rise to the complex, integrated experience that we identify as conscious awareness. This emergent property of consciousness aligns with our intuition that consciousness is more than just the activity of individual neurons but a product of the collective dynamics of the brain. However, it should be made clear that the State Space Theory offers substance to the idea that 'consciousness is an emergent phenomenon.' The theory provides a specific mechanism by which this emergence occurs. Through a neurobiological implementation of Takens' Theorem, these systems learn and then reconstruct models of reality through neurobiologically instantiated state space representations of aspects of reality. In this case emergence, sometimes viewed perhaps as handwaving, is grounded with specific mechanistic and computational underpinning.

This theoretical framework also provides insights into phenomena such as binocular rivalry, where conflicting images presented to each eye lead to alternating perceptions rather than a single, stable image. The recurrent, history-dependent processing within delay coordinate embedding engines allows for the dynamic competition and switching between different neural representations, reflecting the brain's response to conflicting inputs.

In practical terms, this theory suggests that any attempt to alter consciousness, such as through anesthesia or neurostimulation, must account for the nonlinear dynamics of these systems. Disrupting the timing or connectivity within these recurrent networks can lead to significant changes in conscious experience, as seen in the effects of various anesthetics and psychoactive substances. The nonlinear dynamics also imply that small changes in neural activity can sometimes lead to disproportionately large changes in conscious experience.

A discussion of the nonlinear dynamical properties of the brain would be incomplete without a discussion of brain systems existing at the so-called edge of criticality. It has been demonstrated at both the macro- and micro-scale that the brain exhibits behaviors characteristic of certain types of chaotic systems; namely that the brain obeys power law relationships, scale invariance, and self-organized criticality.(Werner 2009) Destabilization of these features is associated with general anesthesia.(Eisen et al. 2024) These features support the presence of nonlinear dynamics in the brain (self-evident even from the examination of the functioning of a single neuron) but more importantly are the macro features that likely underpin the ways in which brain development supports and gives rise, through self-organization, to the heuristic, generic information processing hypothesized in this paper. Indeed, it has been hypothesized (and early work has shown) that the systems operating at the edge of chaos are optimized for adaptability and computational power.(Beggs 2008)

**The Doerig et al Criteria**

As mentioned in the introduction, the criteria laid out by Doerig and colleagues provides a valuable starting point for evaluating the State Space Theory of consciousness. The authors were motivated by a desire to develop a method to compare theories of consciousness, to stimulate discussion and debate with respect to the relative explanatory power of various theories, and to provide a framework for understanding how the predictions of a given theory could potentially be falsified experimentally.

*Paradigm Cases*

It has already been outlined the ways in which the State Space Theory addresses specific paradigm cases of consciousness: cortical blindness, general anesthesia, and ambiguous figures, to name a few. The State Space Theory also accounts for sleep and in particular dreamless sleep. Under the State Space Theory, wakefulness and dreaming states would be accounted for by integrative recurrent processing through delay coordinate embedding engines. By contrast, dreamless sleep would be accounted for through brain activity that subserves known functions of dreamless sleep including clearance of neuroprotective clearance of metabolic waste and memory formation.(Eugene & Masiak 2015; Rasch & Born 2013) In particular, the fact that deep sleep is associated with slow wave states is strongly consistent with the State Space Theory approach; under the State Space Theory, higher-frequency brain activity is requisite for dendritic processing giving rise to delay coordinate embedded reconstruction of state spaces; slow wave activity would be too slow for dendritic integration to give rise to reconstructed state spaces. This is similar to the aforementioned common final mechanism for general anesthesia as proposed under the State Space Theory.

*Unfolding*

The unfolding argument has likewise already been touched upon. Perhaps the most powerful argument against integrated information theory (IIT) and other causal models of consciousness is the unfolding argument.(Doerig et al. 2019) This argument posits that any recurrent neural network (RNN) can be unfolded into a feedforward neural network (FNN) that implements the same input-output function. This argument makes use of the fact that both RNN and FNN are universal function approximators in order to make this claim. On this view, recurrent causal structures are neither necessary nor sufficient to explain consciousness, as any input-output function implemented in RNN can be implemented in FNN.

The State Space Theory (SST) provides a robust rebuttal to this argument in several ways. However, in order to provide a detailed rebuttal, we need to go all the way back to the definition of an algorithm. Succinctly, Evans describes it thus:

> Computer science is the study of information processes… A procedure is a description of a process. A simple process can be described just by listing the steps. The list of steps is the procedure; the act of following them is the process. A procedure that can be followed without any thought is called a mechanical procedure. An algorithm is a mechanical procedure that is guaranteed to eventually finish. (Evans 2011)

Much has been written about Gödel's Incompleteness Theorems, Turing's Halting Problem, and the relationship of these to the nature of human cognition. Sir Roger Penrose, in *The Emperor's New Mind,* famously wrote, "A plausible case can be made that there is an essential *non*-algorithmic ingredient to (conscious) thought processes." (Penrose 1989) While it is beyond the scope of this article to review these considerations in full, in brief, they provide a critical backdrop for understanding the limits of computation and their implications for theories of consciousness, including the State Space Theory. Kurt Gödel's theorems demonstrate that in any sufficiently powerful mathematical system, there are propositions that cannot be proven or disproven within the system. This implies that no formal system can be both complete and consistent. For consciousness, this suggests that the human mind might operate beyond the constraints of formal systems, potentially engaging in non-computable processes. Alan Turing proved that there is no general algorithm that can determine whether any given program will halt or run indefinitely. This highlights a fundamental limit of algorithmic computation, suggesting that some aspects of human cognition might also be non-algorithmic, as proposed by Penrose and others.

However, if we return to the definition of an algorithm (a process that must terminate), one can write a non-algorithmic program to describe human consciousness:

```
while (System.Alive) {be_conscious();}
```

This trivial and rather unsatisfying pseudocode represents a non-algorithmic process; it will not terminate while the interesting case of System.Alive is true. Perhaps we can update it to be slightly more interesting:

```
while (System.Alive) {
switch(ARAS_Mode){
    case ARAS_Modes.Mode_Awake: be_conscious(); break;
    case ARAS_Modes.Mode_REM: be_less_conscious(); break;
    case ARAS_Modes.Mode_NREM: break;
    }
```

These are, of course, frivolous examples of non-algorithmic processes. And if `be_conscious()` *is* an algorithm, a defined set of input-output steps that is continuously returned to in a loop, then we are back to the Godel-Turing-Penrose conundrum. The State Space

Theory goes further in defining consciousness as a non-algorithmic process, and therefore not unfoldable, in two distinct ways.

First, a crucial aspect of the theory is that delay coordinate embedding is not an algorithm but rather a process. Delay coordinate embedding involves creating state space representations through integration of temporal sequences of data. It is only within the dynamical evolution of the system that representations of reality can be instantiated. As a process that inherently requires recurrent loops to integrate information over time, this process cannot be fully replicated by static FNNs as there is no process in an FNN where data can be returned to an earlier point in the network to modulate new information entering this system. . The dynamic reconstruction of state spaces allows for the integration of past data into current processing, emphasizing the role of temporal dynamics in generating conscious experience. Because the theory views the neuronal processes giving rise to consciousness as *processes* rather than as *input-output algorithms*, the unfolding argument is obviated.

Moreover, the State Space Theory posits that the brain's recurrent networks are continuously evolving, shaped by the history of their inputs. In order for a formally defined input-output RNN to be unfolded to an equivalent FNN, it is a prerequisite that the underlying neural network remain unchanged as the equivalent FNN processes sequential inputs. History-dependent changes to the neural network under the State Space Theory - and as supported by abundant neurobiological evidence of synaptic plasticity - again undercuts the argument that brain-based recurrent neural networks can be unfolded to an equivalent FNN without losing essential aspects of their function. Continuous changes in the underlying neural network result in a system that behaves in a non-algorithmic manner.

*Small and Large Network Arguments*

The State Space Theory inherently addresses the small and large network arguments. With respect to small networks, the theory posits that consciousness arises from the integration within higher-order delay coordinate embedding engines. This requires complex, large-scale recurrent networks. Simple networks lack the necessary complexity and hierarchical structure to implement the delay coordinate embedded reconstruction processes fully. With respect to the large network argument, again, the State Space Theory specifically identifies that it is the higher-order delay coordinate embedding engines that give rise to consciousness. While reliant upon the lower order engines in order to provide basic processing, it is the higher-order elements of the network that are associated with consciousness itself. The hierarchical arrangement allows the State Space Theory to explain how consciousness can be both distributed and integrated across the cortex.

*Other Systems Argument*

With respect to the other systems argument, the response of the State Space Theory is straightforward: there is nothing special about the human brain. Any processing system that gives rise to hierarchically arranged delay coordinate embedded representations of the input signals, with a sufficient number of levels in the hierarchy, would give rise to consciousness. Certainly, what is less clear is what constitutes a sufficient number of levels. It is most straightforward to posit under this theory that there is a minimum number of levels below which no there is no conscious experience. Above this number of levels, consciousness would be viewed as arising with increasing richness and diversity of the associated conscious experience.

It is indubitable that even the smallest mammalian brains meet sufficient criteria for some sort of conscious experience.

## Other Theories of Consciousness

A thorough review and comparison to other major theories of consciousness is beyond the scope of this introductory article. That said, it has been pointed out by Doerig and colleagues and others that theories of consciousness have tended to be proffered by their authors in isolation, ignoring the broader context within which they are being offered and failing to compare and contrast the offered theory with existing ideas about the nature of consciousness.(Doerig et al. 2021; Seth & Bayne 2022; Storm et al. 2024) As such, a brief comparison to the major theories of the day is warranted, to highlight alignments, discrepancies, and opportunities for debate and discussion.

### *Higher-Order Theories (HOTs)*

Higher-Order Theories (HOTs) posit that a mental state becomes conscious when it is the target of a higher-order representation, which is a meta-representation of the agent's own mental state.(Seth & Bayne 2022) These theories emphasize the role of anterior cortical regions, especially the prefrontal cortex, in supporting these meta-representations. Both HOTs and the State Space Theory support a hierarchical organization of mental states. The State Space Theory's pseudo-hierarchical arrangement of delay coordinate embedding engines aligns with HOTs' requirement for higher-order representations. Likewise, the State Space Theory's concept of delay coordinate embedding engines creating representations of reality could be seen as a form of meta-representation, though the State Space Theory does not explicitly focus on meta-cognitive states. HOTs emphasize anterior cortical regions, particularly the prefrontal cortex, while the State Space Theory suggests that consciousness arises from the highest-order embedding engines distributed across the cortex, some of which may indeed be commonly localized to anterior cortical regions. In sum, the State Space Theory could be viewed as in many ways providing a mechanistic explanation of HOTs.

### *Global Workspace Theories (GWTs)*

Global Workspace Theories propose that consciousness arises when information is globally broadcasted within a widespread neuronal workspace, particularly involving fronto-parietal regions.(Seth & Bayne 2022) This theory highlights the integration and sharing of information across different brain regions. The State Space Theory's hierarchical delay coordinate embedding engines support the idea of integrating information across different levels, providing a mechanistic underpinning to GWT's global workspace. Both theories recognize the importance of recurrent processing in achieving consciousness. The State Space Theory's recurrent delay coordinate embedding engines align with GWT's recurrent network ignitions. GWTs emphasize a specific neural workspace, often centered on fronto-parietal networks; while the State Space Theory posits a more distributed network involving various delay coordinate embedding engines, again (similar to the discussion on HOTs), there is no aspect of the State Space Theory that would exclude the importance of specific networks for consciousness - with the specific exception that there are likely paradigm cases (such as, for example, the Hogan craniopagus twins) which the State Space Theory would handle better.(Cochrane 2021) The State Space Theory’s detailed computational mechanism of delay coordinate embedding is distinct from GWT's broader concept of global broadcasting.

*Integrated Information Theory (IIT)*

Integrated Information Theory (IIT) links consciousness to the amount of integrated information ($\Phi$) generated by a system. IIT posits that high levels of integrated information, particularly in the posterior cortex, are crucial for consciousness.(Albantakis et al. 2023) Under the State Space Theory, delay coordinate embedding engines create integrated representations of reality, which can be seen as a form of integrated information. The non-linear dynamics and sensitivity to initial conditions in State Space Theory can also align with IIT’s focus on complex information integration. While State Space Theory does not explicitly prioritize posterior cortical areas, the distributed nature of delay coordinate embedding engines can encompass areas that IIT identifies as crucial. The State Space Theory also solves some fundamental problems with IIT, namely the possibility that IIT could ascribe consciousness to entities that are not intuitively conscious; simpler recurrent networks with an insufficient number of levels in the hierarchy would not be expected to give rise to consciousness under the State Space Theory. In the parlance of IIT, we might state that there is a minimum nonzero $\Phi$ associated with conscious experience. As a contrast between the two theories, the State Space Theory does not offer any quantitative measure of consciousness, an advantage that IIT offers. There are likely ways to align the theories that would augment both.

### *Re-Entry and Predictive Processing Theories*

Re-entry and Predictive Processing Theories (RPTs) emphasize the importance of top-down and bottom-up signaling, with a focus on recurrent, re-entrant pathways to enable conscious perception.(Seth & Bayne 2022) The State Space Theory's reliance on recurrent networks for delay coordinate embedding aligns closely with the emphasis on re-entrant pathways in RPTs.

Likewise, the history-dependent development and non-linear dynamics in State Space Theory align with the principles of predictive coding, where past experiences influence current perceptions. RPTs provide additional detail with respect to specific neural pathways in ways that augment the State Space Theory; the State Space Theory offers an advantage here in that it provides a computational mechanism that underlies the predictive models emphasized in RPTs.

## **Philosophical Considerations**

The State Space Theory offers some remarkable advantages with respect to philosophical considerations and quandaries. A brief analysis is offered here; a fuller consideration is again beyond the scope of this manuscript.

*Dualism*

Dualism, the view that mind and body are fundamentally distinct, has long been a contentious issue in the philosophy of mind. State Space Theory, being rooted in physicalist and computational principles, aligns with monist perspectives and seeks to explain consciousness through the interactions and dynamics of physical neural systems. This approach inherently rejects dualism by providing a framework that explains consciousness as an emergent property of physical neural activity. However, due to the *requirement* of a temporal element in the reconstruction of state spaces using delay coordinate embedding, the State Space Theory offers an intriguing answer to the problem of dualist concepts that are, on their face, intuitive. Under the State Space Theory, the mind is, indeed, something different from the brain. A static brain has no mental component, nor does a specific brain state. It is only a brain 'in flight' - actively reconstructing ever more complex state spaces - that gives rise to mental phenomena and conscious experiences. It is a category error to talk of mental states. Consciousness is a verb. Thus, while the mind and brain are not fundamentally different substances as dualism suggests, the State Space Theory introduces a dynamic element to the discussion. Consciousness is not a property of the brain in a fixed state but emerges from the brain's dynamic processes over time.

This dynamic view aligns with process philosophy, which emphasizes becoming over being and sees processes as more fundamental than static entities. The State Space Theory posits that the brain, through its continuous engagement in delay coordinate embedding, generates the flow of conscious experience. The mind, therefore, is the activity of the brain, not a separate entity or a mere collection of brain states. This perspective resolves some dualist intuitions by acknowledging that the brain's dynamic nature is essential for consciousness, thus providing a more nuanced understanding that bridges the gap between physicalist monism and dualist intuitions.

### *The Hard Problem of Consciousness*

The hard problem of consciousness, as formulated by David Chalmers, concerns the question of why and how physical processes in the brain give rise to subjective experiences.(Chalmers 1995) This problem is often seen as a significant challenge for any physicalist theory of consciousness. The State Space Theory posits that consciousness is not just an emergent property of complex, nonlinear dynamical systems, but offers a specific mechanism for that emergence in the form of reconstructed and increasingly abstract and complex state spaces. By explaining how delay coordinate embedding allows for the integration and representation of information in a hierarchical manner, the State Space Theory provides a framework for understanding the emergence of subjective experiences from neural processes. The theory's emphasis on the role of history-dependent development and nonlinear dynamics in shaping individual experiences aligns with the notion that subjective experiences, or qualia, arise from the unique patterns of neural activity shaped by an individual's history and interactions.

### *The Unity of Consciousness*

The unity of consciousness refers to the phenomenon where conscious experiences are integrated into a single, coherent whole, rather than fragmented or disconnected parts. The State

Space Theory explains the unity of consciousness through the hierarchical arrangement of delay coordinate embedding engines. Higher-order delay coordinate embedding engines integrate information from lower levels, creating a unified representation of reality. This hierarchical integration ensures that different aspects of sensory and cognitive processing are combined into a coherent conscious experience. The recurrent nature of neural networks in the State Space Theory further supports the integration and unity of conscious experiences by allowing continuous interaction and feedback between different neural processes.

### *Privacy and Subjectivity of Experience*

One of the key features of conscious experience is its inherently private and subjective nature. Each individual’s experiences are uniquely their own and cannot be directly accessed by others.

The State Space Theory posits that the specifics of how each brain processes information are shaped by an individual’s unique history. This leads to the private and non-shareable nature of conscious experiences, as each conscious entity's neural networks are uniquely configured based on their personal experiences and interactions with the environment.

### *Free Will and Determinism*

The debate between free will and determinism questions whether human actions are determined by prior states and laws of nature or if individuals have the capacity to make free choices. Our intuition that we have free will has been undermined by the conception of the universe as mechanistic and deterministic. This perhaps provides an explanation for the interest in and exploration of quantum theories of consciousness. Quantum mechanical phenomena, in particular probabilistic and stochastic aspects of quantum theory, provide (possibly) a pathway to free will under known physical laws. The State Space Theory of consciousness likewise offers a pathway to free will, but under the auspices of known aspects of nonlinear dynamical systems theory. Under chaos theory, systems may be deterministic and yet unpredictable. In a neural system employing a large number of nonlinear delay coordinate embedding engines, with multiple engines present at the highest level of the hierarchy, there arises the possibility of these systems acting in opposition to one another - multiple 'voices,' modulated by limbic and other signals, presenting different possible behaviors in weighted ways, that represent, to an even higher order engine, possibilities for behavior to select amongst: free will. In future work, we will more rigorously explore these concepts in order to more specifically outline how the State Space Theory can imbue agency to neurobiological systems that meet specific design constraints and the evolutionary advantages implied by this agency.

## Conclusions, Challenges, and Future Directions

In the present work, I have introduced the State Space Theory of consciousness in a comprehensive and yet, in some ways, quite brief manner. This theory, grounded in the physical and computational properties of neural networks, provides a robust framework for understanding the emergence and nature of consciousness. By focusing on delay coordinate embedding as a mechanism for the dynamic and hierarchical integration of information, the State Space Theory offers a novel approach that aligns with many philosophical intuitions while also addressing key empirical and theoretical challenges.

A key next step in the development of the State Space Theory is to explore the ways in which the theory offers specific, falsifiable predictions and gives rise to scientifically verifiable hypothesis testing. We have already seen that the State Space Theory offers a framework by which we can understand the mechanisms underlying general anesthesia and neuroplasticity and how it reconciles specific philosophical quandaries about consciousness. We will explore in future work how the State Space Theory is consistent with other known aspects of neuronal processing - e.g. with machine learning studies on *C. elegans* worms. The State Space Theory makes predictions that could be tested psychometrically. Specifically, there are known quandaries about event related potentials such as the readiness potential that occur prior to consciousness of the associated percept.(Piani et al. 2024) These event related potentials are altered in neurologic and psychiatric disease states.(Sur & Sinha 2009) The State Space Theory offers a window into how recurrent networks - giving rise to coherent, macroscopically observable electroencephalographic features - might be linked back to microscale processing and explain some of these quandaries. Transcranial magnetic stimulation studies could be designed to specifically target and interrupt recurrent processing in specific brain areas to assess for the impact on conscious experience in normal and diseased states, as well as in other altered states of consciousness (e.g. with exposure to psychedelics). These studies could be extended to neuronal microcircuitry using a combination of optogenetics and large scale in vivo neuronal recording, technologies that have only just begun to become available and computationally tractable.

Key challenges for the theory include a number of disparate considerations. One element mentioned earlier is to detail the specific ways in which neurobiological networks are modified to facilitate the learning and access of the myriad state space representations or models of reality. Another challenge is to identify which aspects of behavior and perhaps even which models of reality are inborn. This, a version of the age-old nature/nurture question, leads back to questions about how the neural plan, as implemented in fetal development and as dictated by the plan laid out in DNA and DNA-protein complexes, can specify specific information processing pathways. Presumably, these would give rise to inborn representations of reality that accrued evolutionary survival benefit. It should also be said that any such specification within the neural plan would carry a substantial computational burden on the underlying DNA -- hence, plasticity-mediated genericity to neural computation as the rule rather than as the exception. Another challenge to the theory stands apart: namely, as with any theory in science, that this theory cannot be proven but rather can only withstand the rigors of falsification.

Another critical challenge faced by the State Space Theory relates to how implementation of high-dimensional and multivariable information is managed within the neurobiological system to avoid a number of computational traps and local minima that could be envisioned. Takens' Theorem grounds the theory, at core, as implementation of delay coordinate embedding within recurrent neural network systems. The challenge will be to detail how the vastly simplified Figure 2 is implemented in neural systems with excitatory, inhibitory, and modulatory signaling, with as many as 50,000 synapses to an individual neuron. Another challenge is to envision how state space representations of lower order perceptual phenomena are integrated and bound to form higher order representations, without resulting in excessive sparseness and result in a 'curse of dimensionality' - one of the aforementioned computational traps. This remains to be explored further in future work.

In conclusion, the State Space Theory offers a comprehensive and scientifically grounded explanation of consciousness that addresses many of the philosophical and empirical challenges that have long puzzled researchers. Future work will build on this framework to further explore the mechanisms of delay coordinate embedding, investigate the specific neural pathways involved, and develop empirical tests to validate the theory's predictions. By continuing to refine and expand the State Space Theory, we can advance our understanding of one of the most profound aspects of the human condition and move closer to solving the hard problem of consciousness.

**Figures**

Figure 1: Examples of Delay Coordinate Embedding in the Reconstruction of Dynamical Systems Behavior. (**left**) Velocity time series data. This plot shows the velocity $\bar{v}_x(t)$ of a car moving along a circular racetrack as a function of time, depicted as a sine wave. (**right**) Reconstructed attractor with $m$=2 embedding dimensions: Using delay embedding with an embedding dimension of 2, this plot reconstructs the car's circular motion from the velocity data, revealing a circular attractor. (**bottom**) Lorenz attractor depicted using 3 state variables (x,y,z dimensions), extraction of one of the state variables as a univariate time series, and delay embedding of that univariate time series in three dimensions using time t and a first and second lagged dimension. Figures generated in R and RStudio.

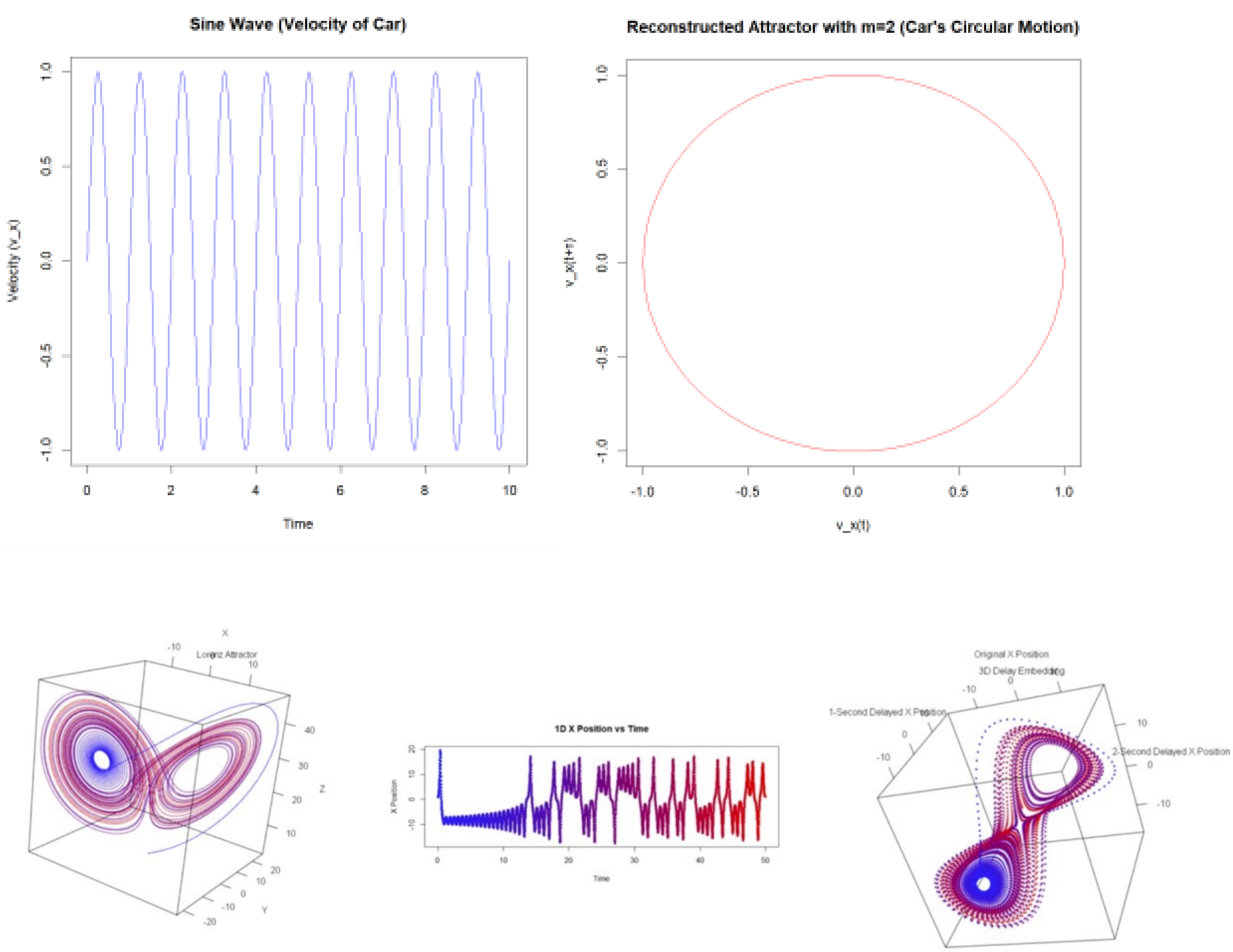


Figure 2: Hierarchical Delay Coordinate Embedding in Neuronal Networks. This figure illustrates the concept of hierarchical delay coordinate embedding within neuronal networks, as proposed by the State Space Theory of consciousness. Sensory input x(t) is received by Neuron 1 and is processed through a series of recurrent loops involving Neurons 2 and 3. The recurrent connections between these neurons create a temporal sequence of embedded states, represented as x(t−1) and x(t−2). These embeddings are integrated over time, enabling the construction of state space representations that capture the dynamics of sensory information. Axon pathways (solid lines) and feedback connections to dendrites (dotted lines) are depicted to emphasize the recurrent nature of the network, essential for delay coordinate embedding and temporal integration of sensory data.

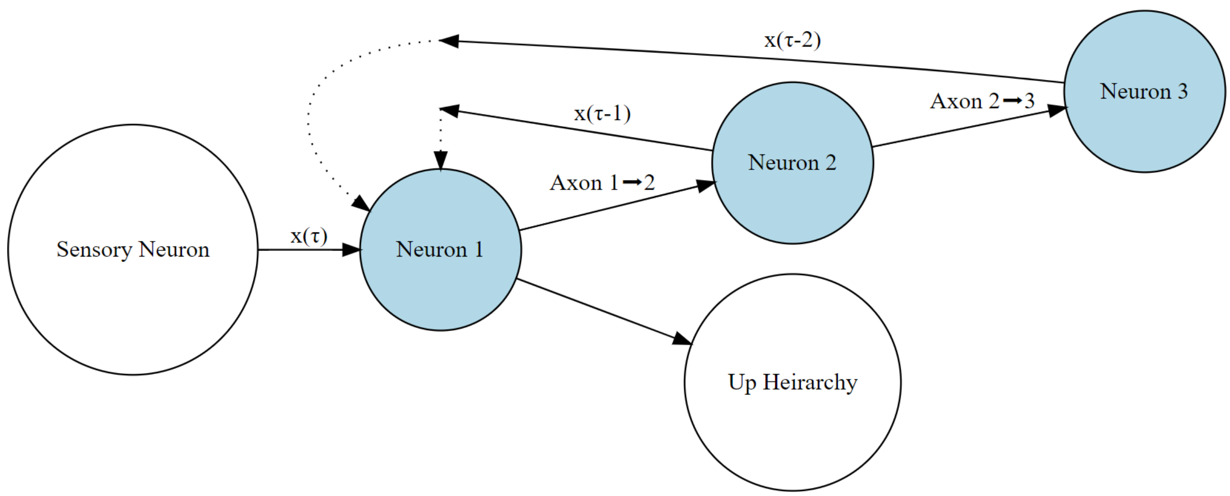


Figure 3: State space representation of color in three dimensions based on receptive cell type in trichromatic (top left and right) and dicromatic (bottom) vision.

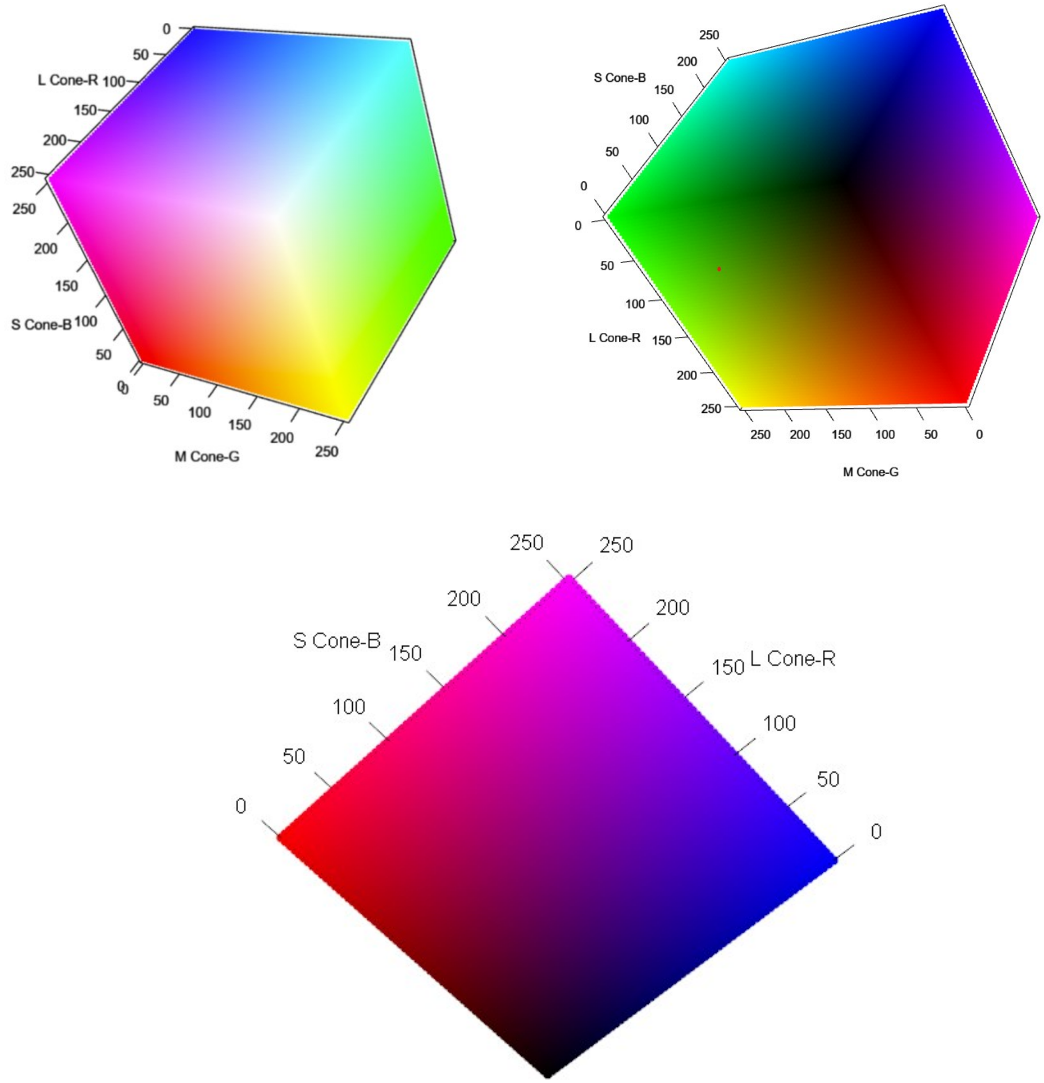
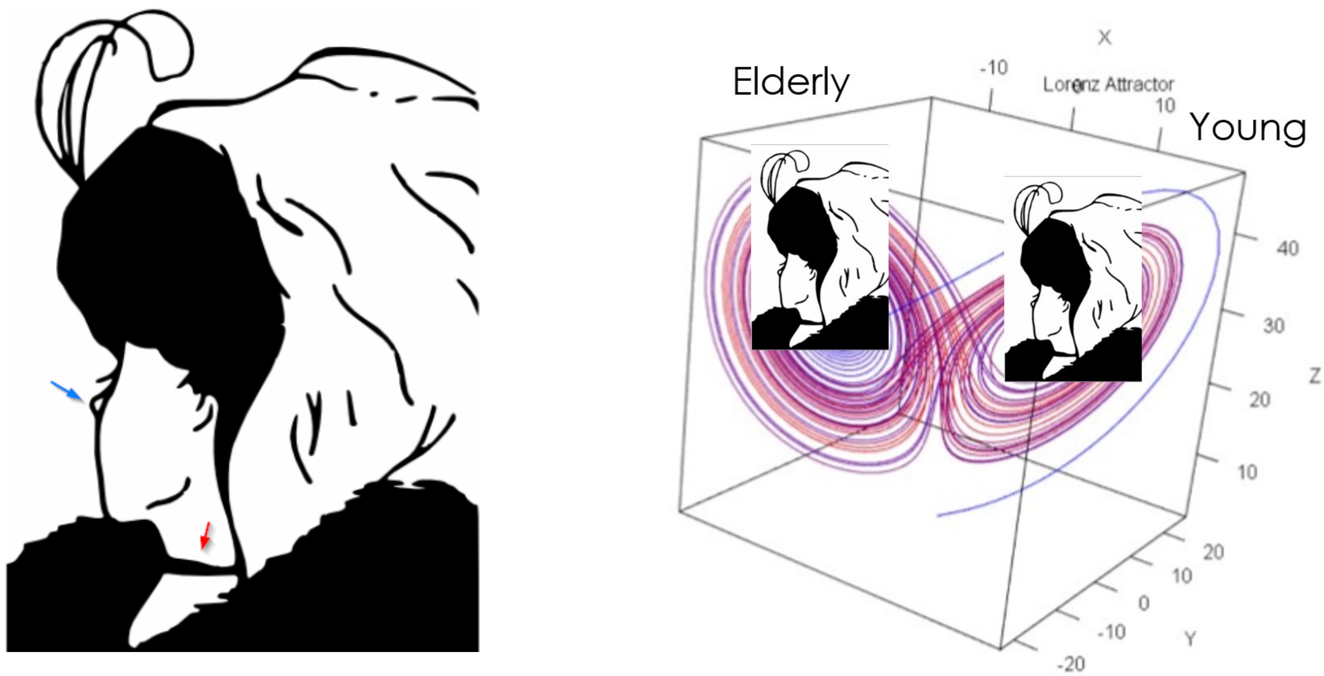


Figure 4: Bistable ambiguous figures through the lens of the State Space Theory. (**left**) The commonly encountered elderly/young woman bistable image marked with a red arrow for the mouth of the elderly woman and a blue arrow for the nose of the young woman. (Public domain images.) (Anon 2023) (**right**) Attractor-based view of the brain favoring one percept or another, based on initial conditions. If the observer focuses on features associated with one stable percept, that percept will dominate.



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